SHEAFY TRACE INDUCES AN ISOMORPHISM ON GLOBAL DERIVED HOMS

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Allowed standard prerequisites (editorial).

- Sheafy trace equation (equation-sheafy-trace, canonical tag 0GLB). cited at statement line 4196. The map whose induced map on global derived homs is claimed to be an isomorphism.
- Global derived hom section (Cohomology on Sites, Section sites-cohomology-section-global-RHom, canonical tag 0B6E). cited at proof lines 4204-4205. The construction expressing RHom as derived global sections of the internal hom.
- Sheafy trace isomorphism on RSheafHom (Lemma lemma-iso-on-RSheafHom, canonical tag 0GLC). cited at proof line 4216. The isomorphism the present lemma is a consequence of.

Source reference. Original equation 0GLB: sources/stacks-chow/tags.txt, source lines 20716–20716. Cited at statement line 4196: the map whose induced map on global derived homs is claimed to be an isomorphism.

Source reference. Original section 0B6E: sources/stacks-chow/tags.txt, source lines 14067–14067. Cited at proof lines 4204-4205: the construction expressing RHom as derived global sections of the internal hom.

Source reference. Original lemma 0GLC: sources/stacks-chow/tags.txt, source lines 20717–20717. Cited at proof line 4216: the isomorphism the present lemma is a consequence of.

Lemma 0.2. *Let $f : X \to Y$ be a separated finite type morphism of quasi-separated and quasi-compact schemes. Let Λ be a torsion ring. For every $K \in D(Y_{étale}, \Lambda)$ and $L \in D(X_{étale}, \Lambda)$ the map (0GLB) induces an isomorphism*

$$R\operatorname{Hom}_X(L, Rf^!K) \longrightarrow R\operatorname{Hom}_Y(Rf_!L, K)$$

of global derived homs.

Proof. By the construction in Cohomology on Sites, Section 0B6E we have

$$R\operatorname{Hom}_X(L, Rf^!K) = R\Gamma(X, R\mathcal{H}om_\Lambda(L, Rf^!K)) = R\Gamma(Y, Rf_*R\mathcal{H}om_\Lambda(L, Rf^!K))$$

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(the second equality by Leray) and

$$R\operatorname{Hom}_Y(Rf_!L, K) = R\Gamma(Y, R\mathcal{H}om_\Lambda(Rf_!L, K))$$

Thus the lemma is a consequence of Lemma 0GLC. □